EXPERIMENT NO:12

Experiment to detect outliers in a given data set.

PROGRAM:

```
#sample calculation for low range(lr) , upper range (ur),percentile
import numpy as np
array=np.random.randint(1,100,16) # randomly generate 16 numbers between 1 to 100
array
```

OUTPUT:

```
array([93, 95, 34, 17, 99, 58,  4, 73, 24, 48, 68, 81, 82, 21, 69,  4])
```

PROGRAM:

```
array.mean()
```

OUTPUT:

```
np.float64(54.375)
```

PROGRAM:

```
np.percentile(array,25)
```

OUTPUT:

```
np.float64(23.25)
```

PROGRAM:

```
np.percentile(array,50)
```

OUTPUT:

```
np.float64(63.0)
```

PROGRAM:

```
np.percentile(array,75)
```

OUTPUT:

```
np.float64(81.25)
```

PROGRAM:

np.percentile(array,100)

OUTPUT:

```
np.float64(99.0)
```

PROGRAM:

```
#outliers detection
def outDetection(array):
sorted(array)
Q1,Q3=np.percentile(array,[25,75])
IQR=Q3-Q1
lr=Q1-(1.5*IQR)
ur=Q3+(1.5*IQR)
return lr,ur
lr,ur=outDetection(array)
lr,ur
```

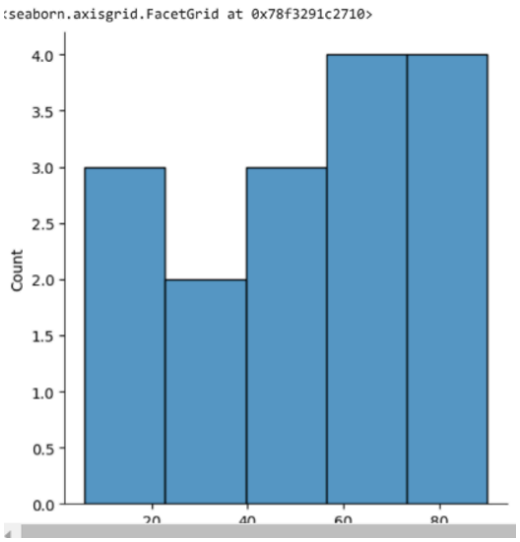
OUTPUT:

```
(np.float64(-63.75), np.float64(168.25))
```

PROGRAM:

```
import seaborn as sns
%matplotlib inline
sns.displot(array)
```

OUTPUT:



PROGRAM:

```
sns.distplot(array)
```

OUTPUT:

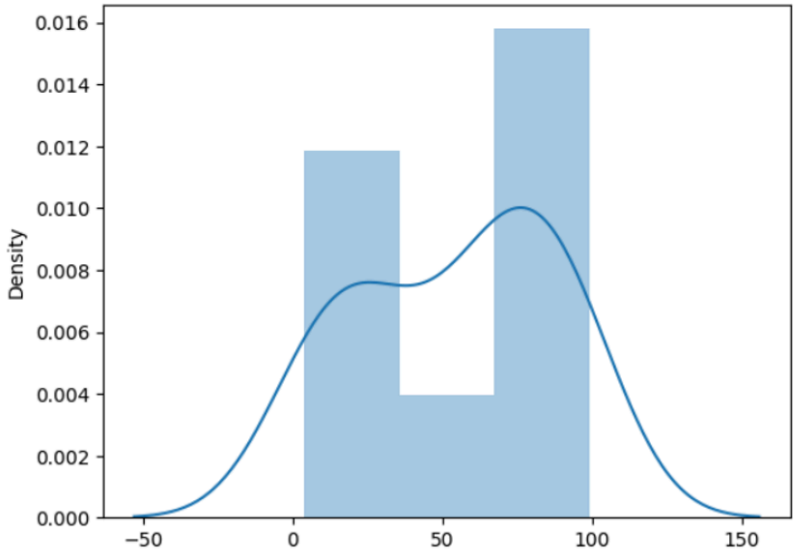
```
/tmp/ipykernel_1091/1133588802.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(array)
<Axes: ylabel='Density'>
```



PROGRAM:

```
new_array=array[(array>lr) & (array<ur)]
new_array
```

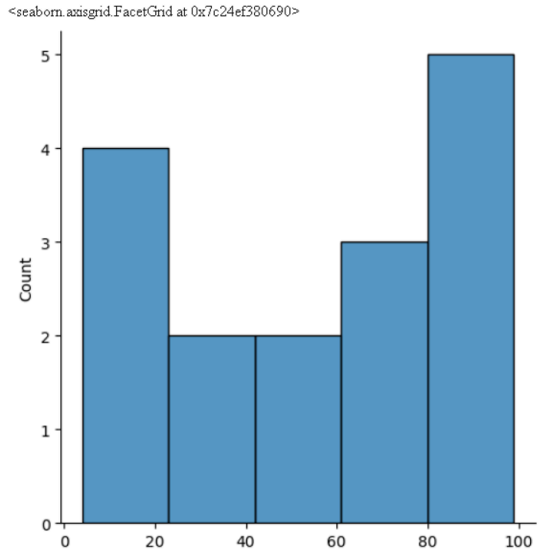
OUTPUT:

```
array([93, 95, 34, 17, 99, 58,  4, 73, 24, 48, 68, 81, 82, 21, 69,  4])
```

PROGRAM:

```
sns.displot(new_array)
```

OUTPUT:



PROGRAM:

```
lr1,ur1=outDetection(new_array)
lr1,ur1
```

OUTPUT:

```
(np.float64(-63.75), np.float64(168.25))
```

PROGRAM:

```
final_array=new_array[(new_array>lr1) & (new_array<ur1)]
final_array
```

OUTPUT:

```
array([93, 95, 34, 17, 99, 58,  4, 73, 24, 48, 68, 81, 82, 21, 69,  4])
```

PROGRAM:

```
sns.distplot(final_array)
```

OUTPUT:

```
/tmp/ipykernel_1091/209491988.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(final_array)
<Axes: ylabel='Density'>
```

